0.1 (1)

In spherical coordinates, the surface element is:

$$dS = R^2 d\Omega = R^2 \sin\theta \, d\theta \, d\phi.$$

Integrating over ϕ, the area of the ring-shaped region within $[\theta, \theta + d\theta]$ is:

$$dS_{\text{ring}} = R^2 \sin\theta \, d\theta \int_0^{2\pi} d\phi = 2\pi R^2 \sin\theta \, d\theta.$$

0.2 (2)

The charge carried by this ring is:

$$dQ_{\text{ring}} = \sigma dS_{\text{ring}} = 2\pi\sigma R^2 \sin\theta \, d\theta.$$

Since the sphere rotates with angular velocity ω, the period of rotation is $T = 2\pi/\omega$. Therefore, the current flowing through this ring is:

$$dI = \frac{dQ_{\text{ring}}}{T} = \frac{2\pi\sigma R^2 \sin\theta \, d\theta}{2\pi/\omega} = \sigma\omega R^2 \sin\theta \, d\theta.$$

0.3 (3)

Each ring forms a current loop with radius $R\sin\theta$. Its magnetic dipole moment is defined as the current multiplied by the area enclosed by the loop:

$$\begin{aligned} dm &= \pi R^2 \sin^2\theta \, dI \\ &= \pi\sigma\omega R^4 \sin^3\theta \, d\theta. \end{aligned}$$

0.4 (4)

Integrating over the entire sphere, we obtain:

$$\begin{aligned} m &= \int_0^{\pi} dm \\ &= \pi\sigma\omega R^4 \int_0^{\pi} \sin^3\theta \, d\theta \\ &= \frac{4}{3}\pi R^4 \sigma\omega \\ &= V R \sigma\omega, \end{aligned}$$

where $V = 4\pi R^3/3$ is the volume of the sphere.

0.5 (5)

The correct diagram is (ウ).

0.6 (6)

Here, $dz > 0$ denotes the positive thickness of the slab, rather than the signed differential of the coordinate z. To distinguish the two, we write the latter as dz_{coord}.

Since $z = R\cos\theta$, we have:

$$dz_{\text{coord}} = -R\sin\theta\, d\theta.$$

Therefore, the thickness of the slab is:

$$dz = |dz_{\text{coord}}| = R\sin\theta\, d\theta.$$

Using the result from (2), we can write:

$$\begin{aligned} dI &= \eta dz \\ &= \eta R\sin\theta\, d\theta \\ &= \sigma\omega R^2\sin\theta\, d\theta. \end{aligned}$$

Comparing both expressions, we obtain:

$$\boxed{\eta = \sigma\omega R}.$$

0.7 (7)

When the sphere is approximated by stacking infinitely thin cylinders, the adjacent end faces of these cylinders cancel each other's magnetic charges, except for the unmatched annular regions at their edges.

The radius of the cross-section at height z is:

$$r_\perp = R\sin\theta.$$

Its infinitesimal change is:

$$dr_\perp = R\cos\theta\, d\theta.$$

Therefore, the area of the unmatched annular region is:

$$dS_{\text{annulus}} = 2\pi r_\perp |dr_\perp| = 2\pi R^2\sin\theta|\cos\theta|\, d\theta.$$

On the other hand, the corresponding spherical surface area is:

$$dS_{\text{sphere}} = 2\pi R^2\sin\theta\, d\theta.$$

Since the magnetic charge density on the end faces of the thin cylinders is $\pm\eta$, the magnitude of the magnetic charge on the unmatched annular region is:

$$|dQ_m| = \eta\, dS_{\text{annulus}}.$$

Thus, the magnitude of the magnetic charge density on the spherical surface is:

$$|\eta_{\text{sphere}}| = \frac{|dQ_m|}{dS_{\text{sphere}}} = \eta|\cos\theta|.$$

The upper hemisphere carries positive magnetic charges, while the lower hemisphere carries negative magnetic charges. Hence, the signed surface density is:

$$\boxed{\eta_{\text{sphere}}(\theta) = \eta\cos\theta}.$$

0.8 (8)

Using the magnetic charge analogy, we can calculate the magnetic field at P by considering the two flat faces of the removed cylinder and the outer spherical surface separately.

From (7), the magnetic surface charge density on the sphere is:

$$\sigma_m(\theta) = \eta \cos\theta.$$

First, consider the two flat faces of the removed cylinder. The lower face carries positive magnetic charges with density $+\eta$, while the upper face carries negative magnetic charges with density $-\eta$.

According to Gauss's law:

$$\oint \mathbf{H} \cdot d\mathbf{S} = \sigma_m S.$$

For a single infinite plane with magnetic charge density σ_m, the magnetic field on either side has a magnitude of $|\sigma_m|/2$. Therefore, the magnetic fields produced by the two faces add up between them:

$$\mathbf{H}_{\text{faces}} = \eta \hat{\mathbf{z}}.$$

Next, consider the magnetic charges distributed on the outer spherical surface.

Since there is no volume magnetic charge and no free current, we have:

$$\nabla \cdot \mathbf{H} = 0, \qquad \nabla \times \mathbf{H} = 0.$$

Therefore, we can introduce a magnetic scalar potential:

$$\mathbf{H} = -\nabla \Phi_m, \qquad \nabla^2 \Phi_m = 0.$$

Since the surface charge density is proportional to $\cos\theta$, we can write the solutions inside and outside the sphere as:

$$\Phi_{\text{in}} = ar\cos\theta, \qquad \Phi_{\text{out}} = \frac{b\cos\theta}{r^2}.$$

At $r = R$, the magnetic scalar potential is continuous:

$$aR = \frac{b}{R^2}.$$

Furthermore, the normal component of the magnetic field has a discontinuity due to the surface magnetic charge:

$$H_{r,\text{out}} - H_{r,\text{in}} = \eta \cos\theta.$$

Substituting the scalar potentials:

$$\frac{2b}{R^3}\cos\theta + a\cos\theta = \eta\cos\theta.$$

Combining these two boundary conditions, we obtain:

$$b = aR^3, \qquad 3a = \eta.$$

Thus, the magnetic field generated by the spherical surface charges inside the sphere is:

$$\begin{aligned}\mathbf{H}_{\text{sphere}} &= -\nabla\Phi_{\text{in}} \\ &= -\frac{\eta}{3}\nabla(r\cos\theta) \\ &= -\frac{\eta}{3}\hat{\mathbf{z}}.\end{aligned}$$

By superposition:

$$\begin{aligned}\mathbf{H}(P) &= \mathbf{H}_{\text{faces}} + \mathbf{H}_{\text{sphere}} \\ &= \frac{2\eta}{3}\hat{\mathbf{z}}.\end{aligned}$$

Since P is located in the vacuum gap left by the removed cylinder, $\mathbf{B} = \mu_0\mathbf{H}$. Therefore:

$$\boxed{B = \frac{2}{3}\mu_0\eta = \frac{2}{3}\mu_0\sigma\omega R}.$$